

Session 4: Reproducible Documents

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esc-r makes a raw cell, just plain text (markdown and code are executed)

1 is a heading, \$ equation \$\$ equation on seperate line

2 Rocket Equation

At time t_0 to rocket starts to expel gas at a *constant mass flow rate* R measured in kg/s and *exhaust velocity relative to the rocket* v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \quad (1)$$

Integrating both sides of Equation 2 from 0 to T we get

3 Rocket Equation

At time t_0 to rocket starts to expel gas at a *constant mass flow rate* R measured in kg/s and *exhaust velocity relative to the rocket* v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \quad (2)$$

Integrating both sides of Equation 2 from 0 to T we get

Open data in python: (Reading a Data File)

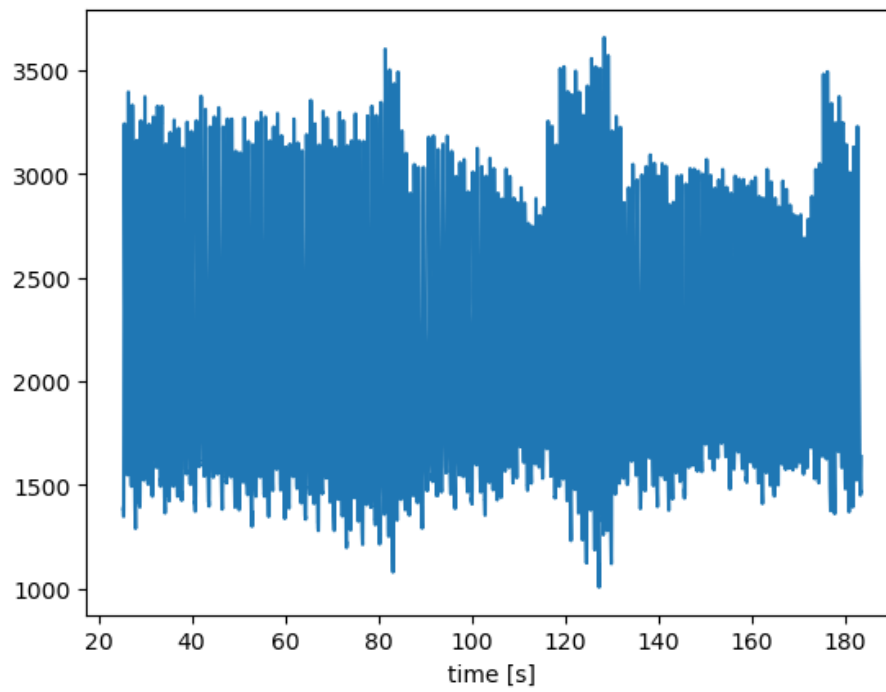
create an empty list:

Numpy is on the Charité server, for calculating things like matlab; matplotlib is another library, pyplot is a collection within

Homework session 3 removed

`Text(0.5, 0, 'time [s]')`

(a) Measurements from an oxymetry sensor as a function of time for 3 cycles



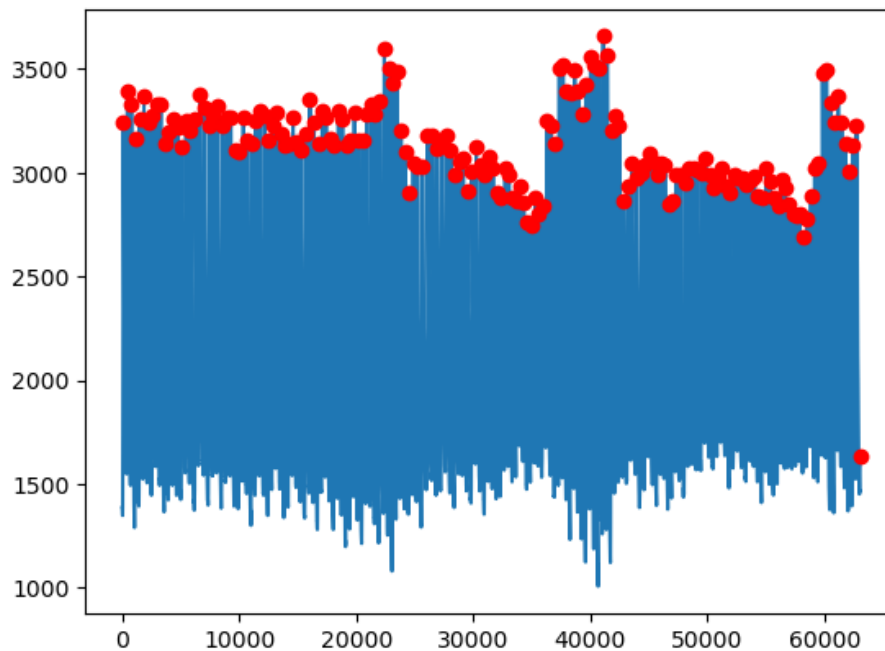
(b)

Figure 1

```

(10,)
dtype('int64')
80
(2, 3)
array([[1, 4],
       [2, 5],
       [3, 6]])
array([[17, 22, 27],
       [22, 29, 36],
       [27, 36, 45]])
np.int64(385)
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
np.int64(10)
np.int64(9)
for i in range (data) #i becomes each point start = max(i-w,0) #its an either or
end = i+w min(i+w,len(data))
window-size w
[96, 438, 774, 1142, 1502, 1851, 2211, 2578, 2929, 3291, 3651, 4003, 4356, 4724, 5092, 545

```



```

array([5, 6, 7, 8, 9])
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19])

```

Homework: how to compute the HR $HR = 60 / \Delta T_{\text{peaks}}$ $tp = \text{time}[\text{peaks}]$
 $tp[1:] - tp[:-1]$

